

OJAS GRAMOPADHYE

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B.Tech in CSE (Hons)
IIT Bombay (2019 - 23)



Incoming MS CS student
NYU Tandon (2024)

WORK EXPERIENCE

Detection and Mitigation of Hallucination in Generative Models

Project Research Assistant

Summer 2023

IBM Research × IIT Bombay

- Researched ways to tackle the effects of **hallucinations in generative** and **large language models** for generalised multi-modal domain setup, and improve abstract reasoning, targeting the field of **medicine and diagnosis**
- Created a **first-of-its-kind** dataset for **Subjective Medical Question-Answering** and proposed **baselines**
- Implemented and evaluated **intermediate pretraining** and **multi-task finetuning** to improve upon model's reasoning skills and tackle **Catastrophic Forgetting**, achieving improved **RougeL** and **BERTScore**
- Extensively experimented with **In-Context Learning** strategies using **Chain-of-Thought**, by using modified **5-shot-codex-COT** and a **custom prompt** to implement a **forward-backward** chaining strategy
- Implemented a **Reasoning Verifier** system on top of the language model by creating a **reinforcement learning reward dataset** to train a **reward model** and apply **elimination** on the **forward outputs** to get the best answer

Japanese OCR Pipeline

Data Science Intern

Summer 2022

Daikin Japan

- Researched and developed an **end-to-end OCR** system capable of text detection and information extraction
- Enhanced the limited Japanese dataset using **document-centric** image processing and data augmentation techniques
- Trained model based on **MobileNetV3 backbone** using **DB algorithm** for text detection to get **F-score** of **0.9**
- Finetuned a **CRNN model** for real-time recognition and achieved a modified **Levenshtein distance** of **0.91**
- Analysed methods for information extraction including Graph Convolution Networks and Spatially-aware transformers.

Locust Occurrence Modelling and Prediction

Data Science Associate | Data Science Intern

Summer 2021

Dtime.ai

- Worked on an **end-to-end** machine learning project to model and forecast **locust outbreaks** in East-African region
- Implemented machine learning algorithms to **extract and engineer new features** and capture the **spatial and temporal** characteristics of the problem, and tackle the heavily skewed nature of the problem, (*as high as 1:5k*)
- Implemented **Bi-directional ConvLSTM** based **U-Net** model for pixel level classification on **67k grid-IDs** of $10km^2$ each, projected over spatial images corresponding to **450+ timestamps**, with respective segmentation masks
- Used a combination of algorithms like **SMOTEENN** and **SMOTETomek**, with **Balanced Ensemble Classifiers** to reduce the effect of class asymmetry and predict a locust occurrence, **minimizing false negatives** successfully

KEY PROJECTS

Spoof-Resistant Face Recognition

Institute Technical Summer Project

Summer 2020

WnCC, IIT Bombay

- Developed a system to differentiate between live and digitally created faces from input image and identify them.
- Built a **convolutional neural network** and trained it on a **self-made** live and fake images dataset
- Extracted **128-D facial encoding vectors** from **dlib's 5-point facial landmark model** to train a support vector machine and work with the **liveness detection model** built using **Tensorflow**, to identify the face
- Implemented **OpenCV's Caffe** based face detector to narrow down ROI, and detect a single face in the image

Lossless High-Entropy Compression Algorithm

Seasons of Code

Summer 2020

WnCC, IIT Bombay

- Achieved the task to convert numeric sequences of a fixed length to compressed versions minimizing the complexity of their **fourier-transform**, achieving **high entropy data compression**.
- Successfully implemented a **seq2seq recurrent neural network** model using **Long Short Term Memory (LSTM)** units with other algorithms, in **Tensorflow** to output compressed signals for efficient transmission and storage
- Achieved the task to convert fixed length numeric signals to compressed versions minimizing their **fourier-transform** complexity, and **recreate the original signal**, using a one-to-one mapping generated using the compressed signal

Quantum Information and Computing

Summer of Science

Summer 2020

MnP Club, IIT Bombay

- Completed a reading project on mathematical aspects of **Quantum Information and Computing**.
- Studied differences in the methods of computation in classical and quantum computers.
- Learned about some basic **quantum algorithms** and the computational ease offered by them over classical counterparts for some tasks known to be computationally complex for classical computers.

Indic Handwriting Recognition

Prof. Ganesh Ramakrishnan — (RnD project)

Spring 2022

IIT Bombay

- Worked on an RnD project trying to model and solve the problem of **Handwriting Recognition** for **Indic languages**
- Implemented a model using convolutional and **Bidirectional-LSTM** layers and **CTCloss** for English language
- Studied and touched some topics like subset selection, knowledge distillation through various research papers

Red Plag : Plagiarism Checker

Prof. Amitabha Sanyal — (Course project)

Autumn 2020

IIT Bombay

- Developed a **plagiarism checker** to measure pairwise similarity between a collection of text files.
- Adopted an algorithm that involved using **separate tokenizers** for languages **C++**, **Java**, **Python** followed by **winnowing of vectors** from **hashed k-grams** to compute similarity percentage.
- Created a web **front-end** using **Angular** framework to visualize results in a graphical manner, and **Django** based web-framework for the **backend** server to maintain the database, and to link and store query results.
- Added **authentication** using *JSON web-tokens* (**JWT**), to restrict access to authorised individuals only.

Automatic Rule Induction - SPEAR

Prof. Ganesh Ramakrishnan — (B.Tech. project)

Spring 2022

IIT Bombay

- Worked on extending the **SPEAR** library to include automatic rule induction approaches.
- Researched and evaluated ways to help integrate existing methods within the **SPEAR** framework to widen the scope of the library and ensure robust evaluation of weak supervision approaches.

C-like language Compiler

Prof. Uday Khedker — (Course Project)

Spring 2021

IIT Bombay

- Built a compiler that could return **Assembly line** code from a C-like program combining Lex and Yacc.
- Coded a scanner and parser that could produce the **Three Address Code** using the **Abstract Syntax Tree**.
- Added the functionality to produce Register Transfer Language using the previously produced outputs of the program.

POSITIONS OF RESPONSIBILITY

Institute Student Mentorship Program (ISMP) Mentor | IIT Bombay

2022 - 2023

- Selected to a team of 144-members out of 388 applicants via a rigorous interview process and peer reviews.
- Responsible to mentor a group of first year students and help & guide them through various difficulties.

Department Academic Mentorship Program (DAMP) Mentor | IIT Bombay

2021 - 2023

Department of CSE, IIT Bombay

- Appointed as **Department Academic Mentor** for two years running. Selected after a process of rigorous interviews and extensive peer reviews. Responsible to guide students in the Department through academic and personal difficulties.

Institute Technical Summer Project (ITSP) Mentor | IIT Bombay

Summer 2021

- Guided a team of 5 members for a machine learning project involving a facial and speech recognition system.

TECHNICAL SKILLS

Programming:	C++, C, Python, Java, Matlab, VHDL, HTML, CSS, SQLite, Bash, Socket, z3
Development:	Android Studio, Angular, Django, Git, L ^A T _E X, Arduino, CSS, JavaScript
Data Science & ML:	Pandas, Matplotlib, Sklearn, Imblearn, Pytorch, Tensorflow, Keras, Numpy, OpenCV
Software Tools:	Git, Jupyter, Quartus, Wireshark, AutoCAD, SolidWorks

COURSES UNDERTAKEN

Computer Science : Discrete Structures, Data Structures and Algorithms + Lab, Data Analysis and Interpretation, Software Systems Lab, Abstractions and Paradigms for Programming + Lab, Digital Logic and Design + Lab, Design and Analysis of Algorithms, Logic for Computer science, Computer Networks + Lab), Computer Architecture + Lab, Database and Information Systems + Lab, Operating Systems + Lab, Automata Theory, Implementation of Programming Languages + Lab, Artificial Intelligence + Lab, Fundamentals of Digital Image Processing, Foundations of Intelligent and Learning Agents, Automatic Speech Recognition, Computer Graphics.

Miscellaneous : Introduction to Electrical and Electronics Circuits, Linear Algebra, Calculus, Basics of Electricity & Magnetism, Quantum Physics, Engineering Graphics & Drawing, Introduction to Public Health Informatics

EXTRACURRICULAR

- Competed in **36th Inter-IIT Aquatics** meet as part of **Men's Aquatics Contingent** which won **2nd position** (2022)
- Competed in **All India IPSC Swimming Championship** (U-19) and bagged a **silver medal** in an event (2016)
- Joined the **62nd National School Swimming Championship** (U-19) organised by **SGFI** (2016)
- Helped organize various Lectures at **Techfest**, Asia's largest science and technology festival. (2019 - 2020)